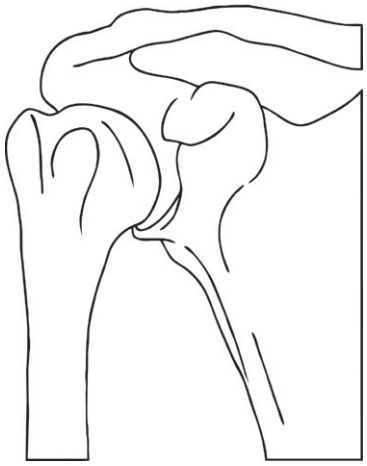
Name:	Class:	Date given:
		Date due in:

Skeletal and **Muscular Systems**

1. Describe the functions of the skeletal system. [3]

2. What is the name given to the part of the skeleton where two bones are linked together? [1]



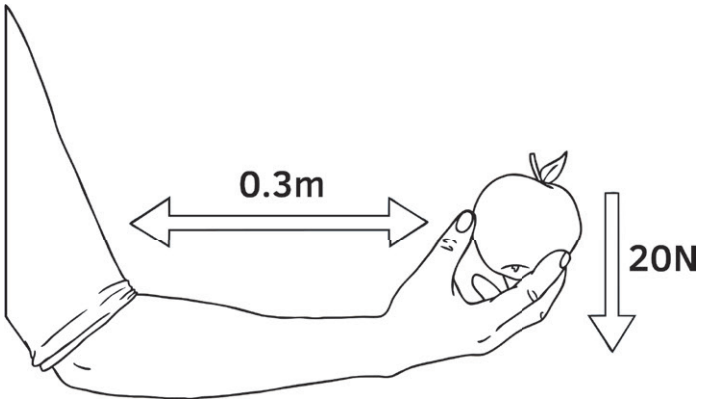
3. Name the type of joint shown in the diagram above and state a part of the body where this type of joint is found. [2]

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4. Use the diagram above to explain how the muscles make the arm move. [4]

5. Muscles exert a turning force. What is the name of this force? [1]



6. Look at the diagram above. Calculate the force exerted. Show all your working out. [3]

Force = _____

Learning Outcomes (tick if achieved)

Q1	I can describe the structure of the human skeleton	
Q2,3	I can describe and explain the biomechanics of the skeleton	
Q4	I can describe the function of the muscles	
Q5,6	I can calculate forces exerted by muscles	

Skeletal and Muscular Systems **Answers**

1. Describe the functions of the skeletal system. [3]

   1 mark for each:

   Support (the body)

   Protect (the organs)

   Movement

   Make blood cells / in the bone marrow

2. What is the name given to the part of the skeleton where two bones are linked together? [1]

   joint

3. Name the type of joint shown in the diagram above and state a part of the body where this type of joint is found. [2]

   Ball and socket joint [1 mark]

   Hip / shoulder [1 mark]

4. Use the diagram above to explain how the muscles make the arm move. [4]

   1 mark for each:

   Antagonistic pair of muscles

   Muscles only pull (so work in pairs)

   Biceps and triceps (may be labelled on diagram)

   2 marks for:

   Biceps contract and triceps relax / arm moves upwards

   Triceps contract and biceps relax / arm moves downwards

5. Muscles exert a turning force. What is the name of this force? [1]

   moment

6. Look at the diagram above. Calculate the force exerted. Show all your working out. [3]

   Moment = Force × perpendicular distance (M = F × d)

   = 20 × 0.3

   = 6 Nm (max 2 marks for answer only with correct units) Force = **6 Nm**